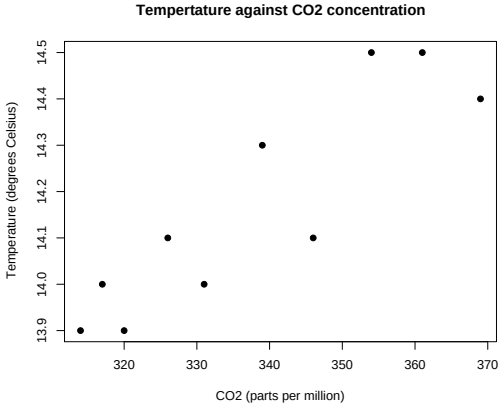
Tutorial 8 - Solutions

1. (a) The scatter plot is shown below. It suggests that there is a positive linear association between CO_2 concentration and temperature.



(b) For these data we have: $\sum x_i = 3377, \sum x_i^2 = 1143757, \sum y_i = 141.7, \sum y_i^2 = 2008.39, \sum x_i y_i = 47888.6$. These give:

$$\begin{aligned} s_{xx} &= \sum x_i^2 - (\sum x_i)^2/n = 3344.1 \\ s_{yy} &= \sum y_i^2 - (\sum y_i)^2/n = 0.501 \\ s_{xy} &= \sum x_i y_i - (\sum x_i \sum y_i)/n = 36.51 \end{aligned}$$

and finally

$$r = \frac{s_{xy}}{\sqrt{s_{xx} s_{yy}}} = 0.892.$$

(c) We want to test $H_0 : \rho = 0$ *v* $H_1 : \rho \neq 0$. Using Fisher's transformation statistic, we have that under H_0

$$\frac{\frac{1}{2}\log\left(\frac{1+r}{1-r}\right)}{\sqrt{1/(n-3)}} = \frac{\sqrt{7}}{2}\log\left(\frac{1.892}{0.108}\right) = 3.79$$

should be a value from the $\mathrm{N}(0,1)$ distribution. This gives p-value $= 2\mathbb{P}(Z \geq 3.79) \approx 0.0002$, so there is strong evidence against H_0 and we conclude that CO_2 concentration and temperature are linearly correlated.

(d) We need to be cautious with the interpretation of this result. The sample correlation coefficient and the hypothesis test show that there is *statistical association* between CO_2 concentration and temperature, but doesn't prove that the *cause* of temperature increase is the increase in CO_2 concentration. Proving this would require specifically designed studies and/or other scientific experiments, and sceptics might argue that other factors (that haven't been accounted for here) are also involved in climate change (although there is now increasing evidence that CO_2 is one of these factors).

2. (a) For this data we have

$$\sum x = 56.8, \sum x^2 = 356.4$$
$$\sum y = 52.74, \sum y^2 = 306.786$$
$$\sum xy = 329.364, n = 10.$$

From here we can calculate

$$s_{xx} = 356.4 - 56.8^2/10 = 33.776,$$
$$s_{yy} = 306.786 - 52.74^2/10 = 28.635,$$
$$s_{xy} = 329.364 - (56.8)(52.74)/10 = 29.8.$$

Therefore

$$r = \frac{s_{xy}}{\sqrt{s_{xx}s_{yy}}} = \frac{29.8}{\sqrt{(28.635)(33.776)}} = 0.958.$$

This shows that there is a strong positive linear relationship.

(b) Test:

$$H_0 : \rho = 0.8 \quad \text{vs} \quad \rho > 0.8.$$

We use the Fisher transformation

$$\frac{1}{2}\log\left(\frac{1+r}{1-r}\right) \sim \mathrm{N}\left(\frac{1}{2}\log\left(\frac{1+\rho}{1-\rho}\right), n-3\right).$$

This gives the test statistic:

$$Z(\underline{X}, \underline{Y}; \rho_0) = \sqrt{n-3}\left(\frac{1}{2}\log\left(\frac{1+r}{1-r}\right) - \frac{1}{2}\log\left(\frac{1+\rho_0}{1-\rho_0}\right)\right).$$

Our observed value of the test statistic (using $\rho_0 = 0.8$)

$$Z(\underline{x}, \underline{y}; 0.8) = \sqrt{10-3}\left(\frac{1}{2}\log\left(\frac{1+0.958}{1-0.958}\right) - \frac{1}{2}\log\left(\frac{1+0.8}{1-0.8}\right)\right) = 2.18.$$

Therefore the p-value (using that this is a one-tailed test) is

$$\mathbb{P}(Z \geq 2.18) = 0.015.$$

Here $Z \sim \mathrm{N}(0,1)$ which we use since under the null hypothesis $Z(\underline{X}, \underline{Y}; \rho_0) \sim \mathrm{N}(0,1)$.

Therefore we have some evidence to reject H_0 in favour of the alternative, i.e. we conclude that $\rho > 0.8$.

3. The likelihood function is given by

$$\begin{aligned} L(\alpha, \beta; \underline{y}) &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{(y_i - \alpha - \beta x_i)^2}{2\sigma^2}\right\} \\ &= (2\pi\sigma^2)^{-n/2} \exp\left\{-\sum_i \frac{(y_i - \alpha - \beta x_i)^2}{2\sigma^2}\right\} \end{aligned}$$

and the log-likelihood is

$$l = l(\alpha, \beta; \underline{y}) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_i (y_i - \alpha - \beta x_i)^2.$$

Differentiating with respect to α and β and setting equal to zero we obtain the equations:

$$\begin{aligned} \sum y_i &= \hat{\alpha} n + \hat{\beta} \sum x_i \\ \sum x_i y_i &= \hat{\alpha} \sum x_i + \hat{\beta} \sum x_i^2 \end{aligned}$$

From the first of these we get

$$\hat{\alpha} = \frac{1}{n}\left(\sum y_i - \hat{\beta} \sum x_i\right) = \bar{y} - \hat{\beta}\bar{x}$$

or, equivalently $\hat{\alpha} = \hat{y} - \hat{\beta}\bar{x}$. Substituting this into the second gives

$$\sum x_i y_i = \frac{1}{n}\left(\sum y_i - \hat{\beta} \sum x_i\right) \sum x_i + \hat{\beta} \sum x_i^2$$

It follows that

$$\hat{\beta}\left(\sum x_i^2 - \frac{1}{n}\left(\sum x_i\right)^2\right) = \sum x_i y_i - \frac{1}{n}\sum x_i \sum y_i,$$

Writing this as $\hat{\beta} s_{xx} = s_{xy}$, it follows that

$$\hat{\beta} = \frac{s_{xy}}{s_{xx}}.$$

4. Show that $\mathbb{E}(\hat{B}) = \beta$ and $\mathbb{V}\text{ar}(\hat{B}) = \frac{\sigma^2}{s_{xx}}$.

We write

$$\begin{aligned}\hat{B} &= \frac{s_{xY}}{s_{xx}} \\ &= \frac{1}{s_{xx}}\sum(Y_i - \bar{Y})(x_i - \bar{x}) \\ &= \frac{1}{s_{xx}}\left(\sum Y_i(x_i - \bar{x}) - \bar{Y}\sum(x_i - \bar{x})\right) \\ &= \frac{1}{s_{xx}}\sum Y_i(x_i - \bar{x})\end{aligned}$$

Taking the expectation, we obtain

$$\begin{aligned}\mathbb{E}(\hat{B}) &= \frac{1}{s_{xx}}\sum \alpha(x_i - \bar{x}) + \sum \beta x_i(x_i - \bar{x}) \\ &= \frac{\beta}{s_{xx}}\left(\sum x_i^2 - n\bar{x}^2\right) \\ &= \beta.\end{aligned}$$

Also we have that

$$\begin{aligned}\mathbb{V}\text{ar}(\hat{B}) &= \frac{1}{s_{xx}^2}\sum(x_i - \bar{x})^2 var(Y_i) \\ &= \sigma^2\frac{1}{s_{xx}^2}\sum(x_i - \bar{x})^2 \\ &= \frac{\sigma^2}{s_{xx}}\end{aligned}$$

5. (a) Using the values computed in Question 1 we have:

$$\begin{aligned}\hat{\beta} &= \frac{s_{xy}}{s_{xx}} = \frac{36.51}{3344.1} = 0.0109 \\ \hat{\alpha} &= \bar{y} - \hat{\beta}\bar{x} = \frac{141.7}{10} - 0.0109\frac{3377}{10} = 10.4891\end{aligned}$$

and fitted regression line is given as $\hat{y} = 10.4891 + 0.0109x$.

(b) Proportion of total variation explained by the regression model is

$$R^2 = \frac{s_{xy}^2}{s_{xx}s_{yy}} = \frac{36.51^2}{(3344.1)(0.501)} = 0.7956$$

i.e. 79.56%. (Note that this can also be computed as r^2.)

(c) We have

$$\hat{\sigma}^2 = \frac{1}{n-2}\left(s_{yy} - \frac{s_{xy}^2}{s_{xx}}\right) = \frac{1}{8}\left(0.501 - \frac{36.51^2}{3344.1}\right) = 0.012799$$

and under the null hypothesis the value of the test statistic is

$$\frac{\hat{\beta}}{\sqrt{\hat{\sigma}^2/s_{xx}}} = \frac{0.0109}{\sqrt{0.012799/3344.1}} = 5.57$$

which should be a value from a t_8 distribution (assuming a full normal model). The critical values of this distribution at $\alpha = 0.01$ are $t_{8,0.995} = -3.355$ and 3.355, so we have strong evidence against the null hypothesis that the slope of the regression is equal to zero.

(d) The R commands and output are given below. You should be able to find the appropriate quantities in the output and verify the results in parts (a), (b) and (c) above (but notice the slight discrepancies caused by rounding off errors).

```
# First input the data
R>temp = scan("")
13.9 14.0 13.9 14.1 14.0 14.3 14.1 14.5 14.5 14.4

R>co2 = scan("")
314 317 320 326 331 339 346 354 361 369

# Fit the model
R>fit.globalw = lm(temp ~ co2)


# Get the ouput
R>summary(fit.globalw)

Call:
lm(formula = temp ~ co2)

Residuals:
Min       1Q   Median       3Q      Max
-0.16062 -0.09183  0.02237  0.07115  0.15204

Coefficients:
             Estimate   Std. Error t value  Pr(>|t|)
(Intercept) 10.483081   0.661635   15.844   2.52e-07 ***
co2          0.010918   0.001956   5.581    0.000522 ***
---
Signif. codes:  0 *** 0.001 ** 0.01 * 0.05 . 0.1   1

Residual standard error: 0.1131 on 8 degrees of freedom
Multiple R-squared:  0.7956,Adjusted R-squared:  0.7701
F-statistic: 31.14 on 1 and 8 DF,  p-value: 0.0005219
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Exam Style Question

(a) There appears to be a fairly weak positive association between issue size and monthly value traded.

[2 marks]

(b) You are given that

$$\sum x_i = 6559.13, \sum x_i^2 = 1967624, \sum y_i = 219.56, \sum y_i^2 = 2166.724, \sum x_i y_i = 58635.97.$$

Using the statistics provided:

$$s_{xx} = 1967624 - \frac{6559.13^2}{26} = 312924.5$$
$$s_{yy} = 2166.724 - \frac{219.56^2}{26} = 312.6246$$
$$s_{xy} = 58635.97 - \frac{(6559.13)(219.56)}{26} = 3246.636.$$

Now

$$r = \frac{s_{xy}}{\sqrt{s_{xx} s_{yy}}} = 0.3282481.$$

[3 marks]

(c) Perform a statistical test to determine if r is significantly different from 0.

We are testing:

$$H_0 : \rho = 0 \text{ versus } H_1 : \rho \neq 0.$$

We use the test statistic

$$T(\underline{X}, \underline{Y}) = \frac{r(\underline{X}, \underline{Y})\sqrt{n-2}}{\sqrt{1 - r(\underline{X}, \underline{Y})^2}}.$$

Note that under the null hypothesis $T(\underline{X}, \underline{Y}) \sim t_{n-2}$ with 24 degrees of freedom.

Our observed value of the test statistic is

$$T(\underline{x}, \underline{y}) = \frac{r(\underline{x}, \underline{y})\sqrt{n-2}}{\sqrt{1 - r(\underline{x}, \underline{y})^2}} = 1.70.$$

We can calculate a p-value as the probability that $T(\underline{X}, \underline{Y})$ takes a value more extreme than 1.70, i.e.

$$p = \mathbb{P}(|T(\underline{X}, \underline{Y})| \geq 1.70) = 2\mathbb{P}(T(\underline{X}, \underline{Y}) \geq 1.70).$$

Using the statistical tables:

$$p = 2(1 - 0.9490) = 0.102.$$

Therefore there is not significant evidence to reject the conclusion that the correlation is 0.

[3 marks]

[Total 8 marks]